

BG77xA-GL&BG95xA-GL

EMUX Application Note

LPWA Module Series

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About the Document

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1 Introduction

The document defines the Quectel MUX (multiplexer) protocol between UE and TE, and provides examples of how to use the MUX function on Quectel BG77xA-GL and BG95xA-GL modules.

This specific implementation of multiplexer protocol is called EMUX and its purpose is to enable the multiplexer control channel similar to the one in the basic option defined in *3GPP TS 27.010*. The EMUX feature allows maintaining four simultaneous sessions via one physical serial port.

Table 1: List of Virtual Port Channels and Their Mapped Functions

EMUX Virtual Port ID	Windows Host	Linux Host	Function
1	BGXX_AT1	/dev/ttyMUX0	AT/PPP
2	BGXX_AT2	/dev/ttyMUX1	AT
3	BGXX_AT3	/dev/ttyMUX2	AT
4	BGXX_NMEA	/dev/ttyMUX3	NMEA Output

Even though EMUX is based on *3GPP TS 27.010* and uses the same header format, it does not support all its features.

The main differences between EMUX and 3GPP-defined CMUX:

- Unlike the standard CMUX, the EMUX configuration is static.
After EMUX is enabled, the module will automatically reset and activate the virtual channels.
- EMUX does not support control messages.
- Virtual channel flow control is not supported.
- Virtual channel baud rate selection is automatically handled by EMUX host-side driver.

1.1. Applicable Modules

Table 2: Applicable Modules

Module Family	Model
BG77xA-GL	BG770A-GL
	BG772A-GL
	BG773A-GL
BG95xA-GL	BG950A-GL
	BG951A-GL
	BG952A-GL
	BG953A-GL
	BG955A-GL

2 Overview of Quectel Multiplexing System

The multiplexer provides mechanisms for conveying data streams between TE and UE over a single physical port. Quectel multiplexing system creates 4 virtual channels on one physical port to transmit multiple streams of data simultaneously. It enables the physical port to act like four real physical channels for the user application-side. Each virtual channel, except port 4 that is reserved for NMEA output, supports SMS, PPP dialing, etc.

All data from the user application-side are packed into different frames, and the frames consist of the data and protocol fields that clearly indicate channel number, information length, FCS, etc. The frames are transmitted as one stream via the serial port. After arriving at the other peer, the data are unpacked as four streams by the multiplexer protocol stack and transmitted to the user application-side.

Each channel between TE and UE is called a DLC (Data Link Connection) and it is established separately and sequentially during the boot up process.

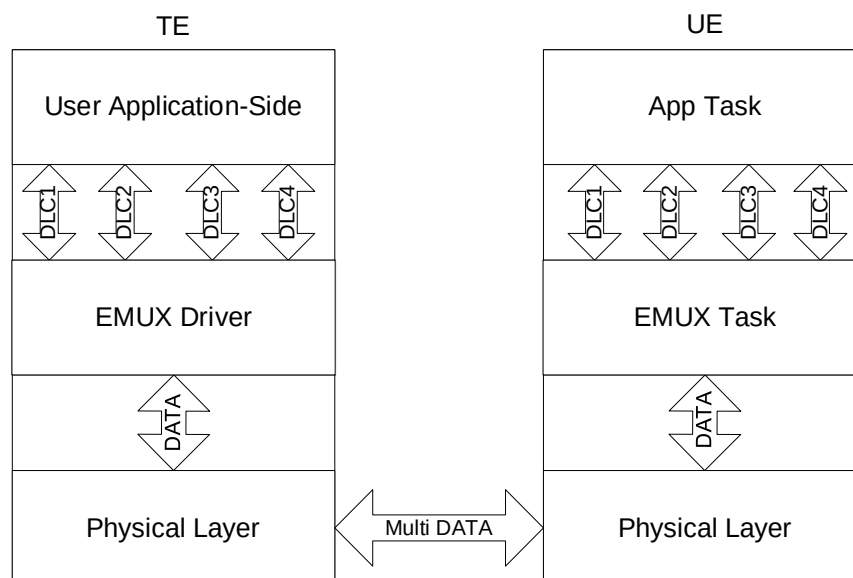


Figure 1: EMUX Architecture

3 EMUX AT Commands

3.1. AT Command Introduction

3.1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

3.1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

Table 3: Types of AT Commands

Command Type	Syntax	Description
Test Command	AT+<cmd>=?	Test the existence of the corresponding command and return information about the type, value, or range of its parameter.
Read Command	AT+<cmd>?	Check the current parameter value of the corresponding command.
Write Command	AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]	Set user-definable parameter value.

Execution Command **AT+<cmd>**

Return a specific information parameter or perform a specific action.

3.2. Declaration of AT Command Examples

The AT command examples in this document are provided to help you learn about the use of the AT commands introduced herein. The examples, however, should not be taken as Quectel's recommendations or suggestions about how to design a program flow or what status to set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there is a correlation among these examples, or that they should be executed in a given sequence.

3.3. AT+EMUX Multiplexing Mode

This command enables/disables the multiplexer channel over the main UART port and sets the channel parameters. After enabling the EMUX feature the user will have four virtual channels at disposal. During the EMUX operation, all URCs will be forwarded to 'BGXX_AT2' (Windows) or '/dev/ttyMUX1' (Linux). This configuration can be changed with **AT+QCFG="emux/urcport"**.

AT+EMUX Multiplexing Mode	
Test Command AT+EMUX=?	Response +EMUX: (list of supported <mode>s),(list of supported <baudrate>s),(list of supported <ifc>s) OK
Read Command AT+EMUX?	Response +EMUX: <mode>,<baudrate>,<ifc> OK
Write Command AT+EMUX=<mode>[,<baudrate>][,<ifc>]	Response OK If there is an error: ERROR
Maximum Response Time	3000 ms
Characteristics	The command takes effect after the module reboots. The configuration is saved.

Parameter

<mode>	Integer type. Multiplexing mode. 0 Disable 1 Enabled
<baudrate>	Integer type. Baud rate of main UART port. Unit: bps. 115200 460800 921600 3000000
<ifc>	Integer type. Specifies if the flow control for EMUX (main UART) port is enabled/disabled. 0 Disable 1 Enable hardware flow control (RTS/CTS)

Example

```

AT+EMUX=1
OK

//The module automatically reboots.

AT+EMUX?
+EMUX: 1,460800,0

OK

```

NOTE

On Quectel's UMTS<E EVB, the maximum value of **<baudrate>** must not exceed 460800 bps.

4 Multiplexer Protocol

This chapter explains the technical details of the multiplexer protocol, including the frame structure and information transfer.

4.1. Frame Structure

All information transmitted between the TE and UE is conveyed in frames.

Table 4: EMUX Frame Structure

Opening Flag	Address	Control	Length	Information	FCS	Closing Flag
1 octet	1 octet	1 octet	1–2 octets	Multiple octets	1 octet	1 octet

4.1.1. Flag Sequence Field

Each frame begins and ends with a flag sequence octet (0xF9).

4.1.2. Address Field

The address field consists of a single octet. It contains the Data Link Connection Identifier (DLCI), the C/R bit and the address field extension bit as shown in the table below.

Table 5: Address Field Format

Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Bit 8
EA	C/R		D	L	C	I	

4.1.2.1. DLCI

The DLCI identifies the virtual channel between TE and UE. Multiple DLCIs can be supported, but the number is implementation-specific. DLCIs are dynamically assigned.

4.1.2.2. C/R

The C/R (command/response) bit identifies the frame as either a command or a response. TE sends a command to the UE with the C/R bit set to 1, and the UE responds with the C/R bit set to 1. The UE sends a command with the C/R bit set to 0, and the TE responds with the C/R bit set to 0.

Table 6: C/R Bit Usage

Command/Response	Direction	C/R Value
Command	TE → UE	1
	UE → TE	0
Response	UE → TE	1
	TE → UE	0

4.1.2.3. EA

The EA bit extends the range of the address field. When the EA bit is set to 1 in an octet, it signifies that this octet is the last octet of the length field. When the EA bit is set to 0, it signifies that it is followed by another octet of the address field.

NOTE

Currently, only EA=1 is supported by the modules.

4.1.3. Control Field

The content of control field defines the frame type. On the modules, only UIH frame is supported. Therefore, the control field is always set to 0.

Table 7: Control Field Coding

Frame Type	HEX (P/F=0)	1	2	3	4	5	6	7	8
UIH (Unnumbered Information with Header check)	0xEF	1	1	1	1	P/F	1	1	1

P/F is the Poll/Final bit. The poll (P) bit set to 1 is used by one station to solicit (poll) a response or a sequence of responses from the other station.

The final (F) bit set to 1 is used by a station to indicate the response frame transmitted as the result of a soliciting (poll) command.

The poll/final (P/F) bit has a function in both command frames and response frames. (In command frames, the P/F bit is referred to as the P bit. In response frames, it is referred to as the F bit.)

NOTE

EMUX only supports data frames (i.e. UIH frames), so the control field is always set to 0.

4.1.4. Length Field

Table 8: Length Field Structure

Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Bit 8
E/A	L1	L2	L3	L4	L5	L6	L7

The L1 to L7 bits indicate the length of the following data field (i.e., the information field) that contains fewer than 128 bytes.

The range of the length field may be extended by use of the EA bit. When the EA bit is set to 1 in an octet, it signifies that this octet is the last octet of the length field. When the EA bit is set to 0, it signifies that a second octet of the length field follows. The total length of the length field is 15 bits in that case.

4.1.5. Information Field

The information field is the payload of the frame and carries the user data and any convergence layer information. The field is octet structured and only presents in UIH frames.

4.1.6. FCS Field

In the case of the UIH frame, the contents of the information field are not included in the FCS calculation. FCS is calculated on the contents of the address, control, and length fields only. This means that only the delivery to the correct DLCI is protected, but not the information.

4.2. Information Transfer

4.2.1. Information Data

In Quectel multiplexing system, information is conveyed using UIH frames. UIH frames may also be used for data in situations where the delays inherent in error-recovery procedures are unacceptable, such as transmission of voice data.

The transmitter takes information from the convergence layer for a particular DLC and places it in the information field of the transmitted frame. Once a UIH frame has been correctly received, the contents of its information field are passed to the convergence layer.

The frames sent by the initiating station have the C/R bit set to 1 and those sent by the responding station have the C/R bit set to 0. Both stations set the P-bit to 0. See **Chapter 4.1.2.2** for more information about the C/R bit.

5 Environment Setup

This chapter describes how to access and use the EMUX virtual channels. Prior to this, EMUX needs to be activated through the main UART port, as described in **Chapter 3.3**.

Windows and Linux are supported host platforms from which the EMUX can be used. A dedicated application that performs virtualization of serial ports on the host side and acts as a proxy between a user and the modem is provided for each platform. After the application is run, virtual channels can be accessed.

5.1. Windows Host

To use the EMUX on Windows OS, the 'com0com' (null-modem emulator) application must be installed on the host computer. After the installation of 'com0com' application, execute the 'eMuxLauncher' application to proceed with the EMUX driver setup. After creating virtual channel bindings, there is no need to repeat this action until their removal. You should enable EMUX on the module with **AT+EMUX** before starting the host application.

```
> eMuxLauncher.bat -help
```

NOTE

Consult Quectel Technical Support (support@quectel.com) for the EMUX driver setup package.

5.2. Linux Host

If there is no binary, the host application needs to be built first using the **make** command.

```
$ make
```

The host application must be run with root privileges, with the following three arguments:

```
$ sudo ./mux /dev/ttyUSB0 /dev/ttyMUX 460800
```

- Physical serial port name (e.g., /dev/ttyUSB0)
- Virtual serial channel prefix (e.g., /dev/ttyMUX)
- Baud rate (e.g., 460800)

5.3. Opening Virtual Channels

Enable EMUX on the module with **AT+EMUX** before starting the host application.

After the host application is started and EMUX is activated, virtual channels can be opened with the preferred application, just like any other serial port, using the preferred terminal application.

Example (Linux Host)

```
$ minicom -D /dev/ttyMUX0 -b 115200
$ minicom -D /dev/ttyMUX1 -b 115200
$ minicom -D /dev/ttyMUX2 -b 115200
$ minicom -D /dev/ttyMUX3 -b 115200
```

6 Appendix References

Table 9: Related Document

Document Name
[1] Quectel_BG77xA-GL&BG95xA-GL_AT_Commands_Manual

Table 10: Terms and Abbreviations

Abbreviation	Description
ABM	Asynchronous Balanced Mode
C/R	Command/Response
DISC	Disconnect
DLC	Data Link Connection
DLCI	Data Link Connection Identifier
DM	Disconnected Mode
FC	Flow Control
FCS	Frame Check Sequence
SABM	Set Asynchronous Balanced Mode
TE	Terminal Equipment
UA	Unnumbered Acknowledgement
UE	User Equipment
UI	Unnumbered Information
UIH	Unnumbered Information with Header Check